Robots Need World Models That Know When the Future Forks: A Goal-Free Probe of Branching Fidelity

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Abstract: Some safety questions are counterfactual: would a slightly different execution of this action, or a slightly different version of the perceived scene, end in a different regime? Following the human intuitive physics engine [1], we answer them with a goal-free *fork probe* that samples the robot's measured execution and perception error, rolls out 64 futures, and alarms when they split into persistent, well-separated groups, with no task labels, failure data or fitted thresholds. In simulated Jenga extraction with real camera perception, the probe tracks ground-truth fragility on three fresh panels (κ 0.70–0.84 against a ceiling of 0.85–0.92, versus 0.32–0.51 when perception error is ignored), including when outcomes are read from image tokens. Used as an evaluation of world models, it shows that learned simulators accurate on average erase forks: a graph-network model halves fork recall, and a stochastic token-space world model reaches only κ 0.18 because its sampled spread is isotropic and ignores where futures branch. Robots need world models for such questions, but only ones with this *branching fidelity*, which current models lack and current benchmarks do not measure.

Keywords: World models, intuitive physics, runtime safety

1 Introduction

Modern manipulation policies, from diffusion policies to vision-language-action models, act by committing to short *action chunks* [2, 3, 4]. Before a chunk runs, a safety monitor can ask whether it is safe. In contact-rich tasks such as Jenga extraction the answer hinges on millimetres: the same command can slide the block out or tip a neighbour, depending on small errors in execution and perception. Such a chunk is precarious even if its nominal rollout succeeds, because uncertainty the robot cannot control decides the outcome.

People judge precariousness in a similar way. In the *intuitive physics engine* (IPE) account [1], observers run a few approximate, noisy simulations over an uncertain scene and call it unstable when the simulated outcomes disagree; noise in both perception and dynamics is needed to explain human judgments [5, 6]. We give a robot the same recipe: perturb the chunk with its *measured* execution error and the scene with its *measured* perception error, roll out 64 futures, and flag the chunk when they split into distinct outcomes. This needs a model the robot can query, and makes the world-model debate concrete: such a model helps only if its rollouts branch where the real dynamics branch.

Contributions. (1) A *fork probe* that validates an action chunk before execution without task labels, failure data or fitted thresholds (Sec. 2). (2) Evidence, with a real camera-based perception model on three fresh test panels, that sampling the robot's own perception error is what makes the judgment match ground truth, including when outcomes are read directly from image tokens

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(Sec. 3.1). (3) *Branching fidelity* as a world-model evaluation: the two learned models we tested predict well on average yet lose the forks; we diagnose why (Sec. 3.2).

2 Problem setup and fork probe

Task and monitor. A simulated Franka Panda (MuJoCo) extracts blocks from Jenga-like stacks. At every chunk start a policy commits to an 8-step action chunk, held for 30 further steps. In our experiments the chunks come from recorded extraction episodes, but the monitor treats the policy as a black box: it sees only the commanded chunk, one camera image and the robot's proprioception, so it applies unchanged to any chunked policy. (How it rolls out futures, with a simulator or a learned model, is the subject of Sec. 3.2.)

What we call unsafe. A chunk is *fragile* if deviations the robot cannot control, namely its execution error and the error in its perception of the scene, can send the outcome into qualitatively different regimes. Three reasons motivate this definition. It is *task- and policy-agnostic*: it needs no goal predicate or failure label, which a monitor around a generalist policy cannot assume. It isolates cases where the robot's *intent no longer determines the outcome*; if all plausible executions agree, the result is a property of the plan. And in contact-rich manipulation damage typically happens at such boundaries, which we verify against physical harm. A chunk that fails *deterministically* is not flagged; that needs a task-level check.

Fork probe. We draw 64 execution-error snippets from the robot's measured tracking residuals (median chunk maximum 3.2 mm), subtract them from the commanded positions, and roll out. The probe alarms if, at hold steps 10 *and* 30, the endings (object positions, or later image tokens) projected on their first principal axis prefer two Gaussians to one by BIC, are well separated (Ashman's $D > 2$ [7]), have at least 2 rollouts in the minority, and keep the same partition. These constants are fixed statistical conventions, not fitted to labels.

Uncertainty-aware probes. The robot does not see the true scene s but an estimate $\hat{s}$: a frozen DINOv2 encoder [8] on one close-camera image plus a keypoint head regresses the poses of the three blocks in play (held-out error 2.5 mm). Its validation residuals define an error model (full 9×9 position covariance plus rotation covariances). The *naive* probe starts every rollout at $\hat{s}$. The *aware* probe (v1) starts rollout j at $\hat{s} + e_j$, with e_j drawn from the error model, as the IPE does. The *Bayesian* probe (v2) draws 5 hypotheses $\hat{s} + d_m$, runs an aware probe around each, and alarms if at least 3 split.

Ground truth and protocol. *Fragility truth* applies the aware recipe around the *true* state with exact physics, so it reflects the robot's uncertainty, not one exact outcome. Two independent seeds (A, B) agree at Cohen's κ 0.85–0.92, the attainable ceiling. Because this truth shares the split test with the monitors, it measures how well a monitor handles perception, not whether splits are dangerous. For that we use an independent physical label: a *harm fork* is a state where 2–62 of 64 truth rollouts topple a neighbouring block. Every confirmation uses a *fresh panel* of unseen stack layouts (up to 60 topple forks plus 100 safe states) under a protocol frozen before the data existed.

3 Results

With exact physics and the true state, the split test alarms on all 23 development topple forks at a 10% alarm rate on quiet states. What remains when the robot relies on its own perception, and then on a learned model instead of the simulator?

3.1 Handling perception error

Real perception errors of 2–4 mm are comparable to execution noise, so the perceived scene is effectively a different scene. The naive probe, which trusts $\hat{s}$, reaches only 38–55% of the ceiling

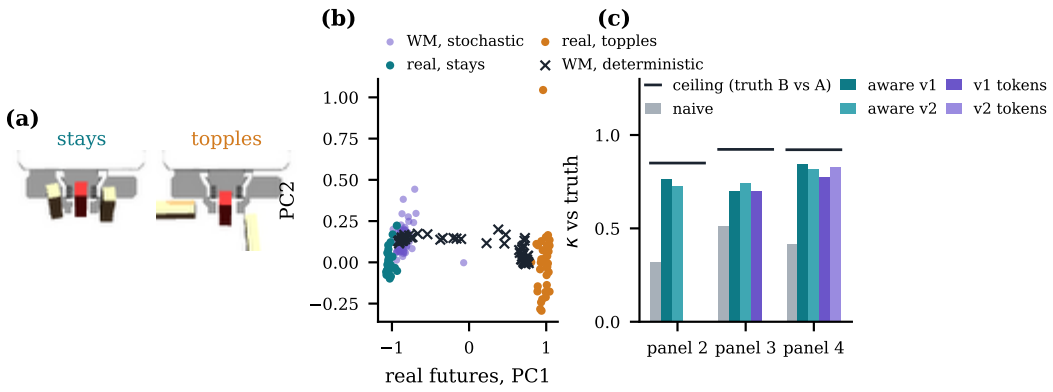


Figure 1: (a) Two of 64 endings of one panel-3 chunk under execution error: the neighbour stays or topples. (b) A held-out fork (most balanced real split): real latent futures (colour = outcome) on their principal axes, with the token world models' futures for the same image and actions; the deterministic model spreads along the branch axis, the stochastic one stays near one outcome with its spread mostly off this plane. (c) Agreement with fragility truth on three fresh panels; lines mark the ceiling.

Table 1: Fork probes with real perception on fresh panels (exact physics). κ: agreement with fragility truth A; harm: recall on harm forks; tokens: outcomes read from image tokens.

	Panel 2 (153 states)		Panel 3 (160)		Panel 4 (156)	
monitor	κ	harm	κ	harm	κ	harm
ceiling (truth B vs A)	0.85	–	0.92	–	0.92	–
naive ($\hat{s}$ only)	0.32	–	0.51	0.65	0.42	0.67
aware v1	0.76	–	0.70	0.77	0.84	0.91
aware v2 (Bayesian)	0.73	0.90	0.74	0.90	0.82	0.89
aware v1, tokens	–	–	0.70	0.77	0.77	0.88
aware v2, tokens	–	–	–	–	**0.83**	**0.88**

κ; sampling the measured perception error raises this to 76–91% (Table 1, Fig. 1c), and the aware probes catch 77–91% of harm forks. This is the IPE's prescription: judge fragility at one's own uncertainty, not at a point estimate.

Reading outcomes from images. A deployed monitor should not need object states to judge outcomes. Applied unchanged to flattened DINOv2 tokens of rendered endings, the split test agrees with the position-based test at κ 0.86–0.90 on aware rollouts; tokens ignore harmless sub-centimetre splits (22 of 24 disagreements on panel 2), acting as a consequence-scale floor. The fully visual Bayesian probe meets its frozen gate on panel 4 (harm recall 0.88, 95% CI 0.77–0.94; 7% alarms on non-harm states). Misses come from where rollouts start, not from reading images: on panel 3, token- and position-read v1 each miss 14 of 62 harm forks, 12 because the single estimate put the rollout cloud on the safe side of the edge and 2 because nearly all rollouts toppled (a deterministic failure). Panels differ in difficulty, so v2's extra caution is not shown to help.

3.2 Branching fidelity of learned world models

So far the rollouts used MuJoCo with privileged geometry. With a learned world model in its place, the probe becomes a test of the model: do its rollouts branch where physics branches? We tested two. A graph-network state model trained with a counterfactual objective ranks fragile states well (AUC 0.97), yet under a refined probe it keeps 31% (8–49% over 10 seeds) of held-out topple forks where exact physics keeps 65.5%. A stochastic token-space world model, with DINOv2 tokens compressed by a fixed per-patch PCA to $28 \times 28 \times 8$ and a flow-matching transition trained on 8 futures per image whose starts were sampled within perception error, also predicts well on average:

Table 2: Token world models: κ, harm on fresh panel 4; diagnostics on 60 held-out states (spread vs real; PC1: share on the main axis at forks; ratio: spread on forks / non-forks).

dynamics	κ	harm	spread	PC1	ratio
reference (MuJoCo + v2 tokens; real futures)	0.83	0.88	1.00	0.57	4.8
token WM, stochastic	0.18	0.18	0.98	0.15	1.3
token WM, deterministic	0.42	0.60	0.04	0.69	32
[TODO: iteration 2]					

both token models beat a no-motion baseline on 19-step latent error by 3–4×. Yet it fails on fresh panel 4 (κ 0.18, harm recall 0.18, Table 2), below its own deterministic control (κ 0.42).

Why the stochastic model fails. It reproduces the *amount* of spread but not its *shape* or *placement*: the spread is isotropic and nearly the same on forks and non-forks, so the model learned the average uncertainty of futures, not where they branch. Other samplers are worse. The deterministic model's spread is tiny but aligned with the branch axis and 32× larger on forks (Fig. 1b); started from rendered perturbed scenes it doubles agreement with real futures (κ 0.17→0.37). As in the IPE, uncertainty should enter through the initial conditions and be amplified by roughly right dynamics, not be added as per-step generative noise. A second iteration (a learned operator perturbing the start latent by physical perception error, then deterministic dynamics) is under evaluation [TODO: result].

4 Related work

Simulation under uncertainty. Beyond the IPE and counterfactual accounts of physical support [9], robotics has long run Monte Carlo rollouts over pose, friction or model parameters to score grasps and pushes [10, 11] or to build safety filters [12, 13]; these judge outcomes with a task predicate or constraint. Zapolsky and Drumwright [14] proposed detecting divergent behaviour from particle traces but evaluated task-failure detection. Learned stability classifiers need labels [15]. We perturb with the robot's own measured errors and judge only the structure of the outcome set, a Monte Carlo test of proximity to a basin boundary [16].

Runtime monitors. Latent safety filters use learned failure classifiers, including on Jenga [17, 18], and failure detectors use policy-side uncertainty calibrated on successes [19, 20]. These measure what a model does not know; the fork probe measures whether the physics is at a branch.

World models and their uncertainty. Learned simulators are accurate on rigid contact [21, 22], and latent world models support planning [23, 24, 25], but stiff contact is hard to learn [26], and audits find poor physical grounding and optimism under failure-inducing actions [27, 28, 29]. Stochastic and ensemble models represent uncertainty [30, 31] but are judged by likelihood or return; we ask whether their spread falls where the dynamics branch.

5 Conclusion and limitations

A robot can judge fragility the way people do, by sampling its own perception and execution uncertainty through a rough simulator. For the question this workshop poses, our results suggest that pre-execution safety checks of this kind need a queryable world model with branching fidelity, which the two learned models we tested lack despite low average error, and that uncertainty belongs in the initial conditions rather than in per-step noise. Average-error benchmarks would rank these models well; we propose branching fidelity as an additional axis.

Limitations. Simulation, one task family; MuJoCo probes use privileged geometry; fragility truth shares the split test (harm is the independent check); single world-model runs.

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